PROJECT DOCUMENTATION FOR THE ASSESSMENT ACTIVITY

Student Performance Analysis System

A Python project built with NumPy and Pandas

1. Project Title

Student Performance Analysis System – a Python project built with NumPy and Pandas.

2. Objective

Turn a plain list of marks into clear results, without any manual calculation.

Teachers usually keep marks in a register or a spreadsheet, and then work out totals, averages, grades and pass or fail by hand. That is slow, and one small slip changes a student's result. It is also not easy to see at a glance who is leading the class or which subject needs more attention.

This project solves that. It reads the marks from a CSV file and calculates everything automatically for each student. It then prints a ranked result list, a class summary, a subject-wise comparison and a list of high scorers. Along the way, it was a chance to practise Python, NumPy and Pandas on a problem that feels real.

Prepared by	Bushra Zahgeer
Programme	B.Tech, 6th Semester
Branch	CSE
College	Government College of Engineering & Technology, Safapora, Ganderbal
Submitted to	Chandini Madam
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3. Technologies Used

Tool	Used for
Python	The language the whole project is written in.
Pandas	Loading the CSV, creating new columns, sorting, filtering and saving the results.
NumPy	Summing and averaging the marks of each student, and calculating the subject averages.
CSV files	For the input data and for the saved output.
Git and GitHub	To keep the project and submit it.
VS Code	For writing and running the code.

4. Dataset Description

The data is stored in `students.csv` and has records of 21 students. Each row is one student.

Column	What it holds
Student_ID	A unique ID for the student (2301 to 2321)
Name	The student's name
Department	CSE, IT or ECE
Python	Marks out of 100
DBMS	Marks out of 100
Networking	Marks out of 100
AI	Marks out of 100
Attendance	Attendance in percent

The program adds four more columns while it runs: `Total_Marks`, `Average_Marks`, `Grade` and `Result`.

5. Implementation

The work is split into two files. `main.py` runs the program from start to end, and `functions.py` holds the helper functions it calls.

1. **Loading the file.** `main.py` first checks that `students.csv` is in the same folder and prints a message if it is not. Otherwise the file is read with `pd.read_csv()`, and the student count, column names and first five rows are shown.
2. **Totals and averages.** `add_totals()` picks the four subject columns, converts them to a NumPy array and sums and averages each row. The average is rounded to two decimal places.
3. **Grade and result.** `get_grade()` converts the average into a grade, and `get_result()` is applied to every row to decide Pass or Fail.

Average	Grade
90 and above	A+
80 and above	A
70 and above	B
60 and above	C
50 and above	D
Below 50	F

> A student passes only when **every subject mark is 40 or more** and **attendance is 40% or more**.

4. **Ranking.** The students are sorted by average marks, highest first, and shown with their total, average, grade, attendance and result.
5. **Class summary.** The program finds the class average, the highest and lowest scorers using `idxmax()` and `idxmin()`, the number of passes and fails, and the pass percentage.
6. **Subject analysis.** `subject_summary()` returns the average, highest and lowest mark of each subject, and the subject with the best class average is printed.
7. **High scorers.** Students with an average of 80 or more are filtered and listed.
8. **Saving.** The full results go to `student_results.csv` and the subject table to `subject_analysis.csv`.

6. Key Features

Feature	What it does
File check	Shows a clear message if students.csv is missing instead of crashing
add_totals()	Adds total and average marks using NumPy
get_grade()	Gives each student a grade from A+ to F
get_result()	Decides Pass or Fail using marks and attendance
subject_summary()	Gives the average, highest and lowest mark of every subject
Ranking	Lists all students from best to lowest average
Class summary	Shows the class average, topper, lowest scorer and pass percentage
High scorers	Lists students with an average of 80 or more
Saved output	Writes the results to two CSV files

7. Output / Screenshots

```
============================================================
STUDENT PERFORMANCE ANALYSIS
============================================================
Total students: 21
Columns: ['Student_ID', 'Name', 'Department', 'Python', 'DBMS', 'Networking', 'AI', 'Attendance']

 Student_ID           Name Department  Python  DBMS  Networking  AI  Attendance
       2301  Mudasir Ahmad        CSE      72    64          65  67          73
       2302    Umar Farooq        ECE      69    68          48  77          80
       2303   Ankit Tiwari         IT      35    60          51  45          75
       2304 Tabassum Akhtar         IT      66    63          77  84          97
       2305  Harsh Vardhan        ECE      79    79          61  73          85
```

```
============================================================
STUDENT WISE RESULTS
============================================================
 Student_ID             Name Department Total_Marks Average_Marks Grade Attendance Result
       2307   Bushra Zahgeer        CSE       382.0         95.50    A+         98   Pass
       2315     Vivek Sharma        ECE       366.0         91.50    A+         81   Pass
       2308   Deepak Chauhan        ECE       345.0         86.25     A         64   Pass
       2313      Kavya Reddy        ECE       327.0         81.75     A         64   Pass
       2319      Meera Joshi         IT       322.0         80.50     A         66   Pass
       2317    Hamza Qureshi        ECE       321.0         80.25     A         86   Pass
       2309        Sana Bhat        CSE       320.0         80.00     A         75   Pass
       2321       Aman Verma        ECE       315.0         78.75     B         97   Pass
       2316      Insha Malik         IT       314.0         78.50     B         67   Pass
       2312      Rajesh Koul         IT       300.0         75.00     B         80   Pass
       2306    Tariq Hussain        ECE       298.0         74.50     B         81   Pass
       2305    Harsh Vardhan        ECE       292.0         73.00     B         85   Pass
       2304  Tabassum Akhtar         IT       290.0         72.50     B         97   Pass
       2310        Saba Wani         IT       276.0         69.00     C         90   Pass
       2301    Mudasir Ahmad        CSE       268.0         67.00     C         73   Pass
       2302      Umar Farooq        ECE       262.0         65.50     C         80   Pass
       2318       Rohit Bhat        CSE       254.0         63.50     C         75   Pass
       2320       Sneha Iyer        CSE       253.0         63.25     C         80   Pass
       2314       Yasir Khan        CSE       235.0         58.75     D         77   Fail
       2311      Rubina Khan        CSE       212.0         53.00     D         76   Fail
       2303     Ankit Tiwari         IT       191.0         47.75     F         75   Fail
============================================================
```

```
============================================================
CLASS SUMMARY
============================================================
Class average: 73.13
Highest: Bushra Zahgeer - 95.5
Lowest: Ankit Tiwari - 47.75
Passed: 18
Failed: 3
Pass percentage: 85.71 %

============================================================
SUBJECT WISE PERFORMANCE
============================================================
   Subject  Average_Marks  Highest_Mark  Lowest_Mark
    Python          73.62            96           35
      DBMS          72.76            94           38
Networking          70.81            97           39
        AI          75.33            95           45
Best subject (by class average): AI

============================================================
TOP STUDENTS (average >= 80)
============================================================
 Student_ID            Name  Average_Marks Grade
       2307  Bushra Zahgeer          95.50    A+
       2315    Vivek Sharma          91.50    A+
       2308  Deepak Chauhan          86.25     A
       2313     Kavya Reddy          81.75     A
       2319     Meera Joshi          80.50     A
       2317  Hamza Qureshi          80.25     A
       2309       Sana Bhat          80.00     A

Saved student_results.csv and subject_analysis.csv
PS C:\Users\zmune\OneDrive\Desktop\Bush>
```

8. Final Outcome

Running `main.py` now gives a complete performance report in a few seconds, and it also saves the processed data so it can be opened in Excel.

For the 21 students in the dataset, the results were:

73.13	85.71%	95.50	47.75
Class average	Pass percentage (18 passed, 3 failed)	Highest average Bushra Zahgeer (A+)	Lowest average

7	AI
Students with an average of 80 or more	Best subject, with a class average of 75.33

The project therefore does what it was built for: it replaces manual calculation with a program that is quick, repeatable and easy to read.

9. Challenges & Learning

- The program stopped when the CSV was not found. I fixed this by checking the file path first and printing a helpful message.
- The pass or fail rule needed several columns at once. I used `apply()` with `axis=1` so the function receives a full row.
- A student could have a good average but still fail. I learned that the rule has to check every subject and attendance, not just the average.

- Keeping the code tidy. I moved the calculations into `functions.py` so `main.py` stays short and readable.

What I learned: reading and handling data with Pandas, calculating with NumPy arrays, sorting and filtering a DataFrame, writing reusable functions, splitting a project across files and putting it on GitHub.

10. GitHub Repository

Repository link: https://github.com/bushrazahgeer/student-performance-analysis